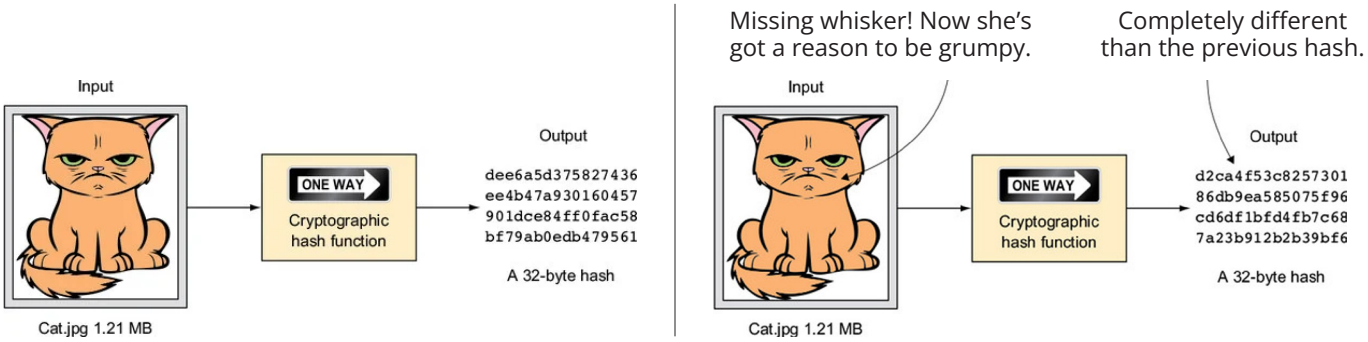


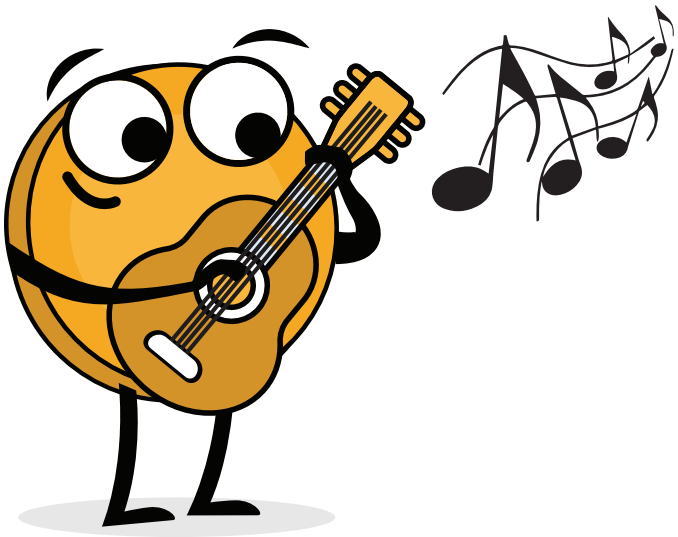
The **output**, or **hash**, is always the same length, no matter how long the original information was.

Bitcoin uses a few specific types of hash **function** called **SHA-256 and RIPEMD160**.
A few examples below:

- Notice that a period in the second input changes the output completely when compared to the first one.
- The third input is a huge file yet the output is still the same fixed length as the other two.

SHA256 hash of the string ***hello world***
B94d27b9934d3e08a52e52d7da7dabfac484efe37a5380ee9088f7ace2efcde9

SHA256 hash of the string **hello world.**
7ddb227315f423250fc67f3be69c544628dffe41752af91c50ae0a9c49faeb87

SHA256 hash of the downloadable iso file **Ubuntu 18.10**
7b9f670c749f797a0f7481d619ce8807edac052c97e1a0df3b130c95efae4765

Hashing can also be thought of as a musical score that captures the essence of a piece of music. Just as a musical score is a unique representation of a tune, a hash value is a unique representation of a piece of data. By comparing the score of a piece of music with the actual performance, a musician can determine if the performance is accurate. Similarly, by comparing the hash value of received data with the original hash value, one can determine if the data has been altered during transmission.